Physical Properties of Matter

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Mass, volume, density, solubility, conductivity, and magnetism are physical properties that can be observed or measured without changing a substance into something new.

Round 1 - Retrieve It

1. What does mass measure?

2. What does volume measure?

3. What is the formula for density?

4. Name three other physical properties on the cue card.

Round 2 - Sketch It

Make a six-part property wheel. Label mass, volume, density, solubility, conductivity, and magnetism. Add one quick test or tool for each.

Physical Properties of Matter - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A rock has mass 54 g and volume 20 mL. Calculate its density.

2. Which property is most useful for separating iron filings from sand?

3. Which property could help compare copper wire and plastic for use in a circuit?

Round 4 - Reason from Evidence

A student says, "If I measure a property, I always change the substance." Use two physical-property examples to show why that is incorrect.

Physical Properties of Matter - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A **Articulate**
Explain Physical Properties of Matter in your own words. Use one example that is not printed on the cue card.

C **Connect**
Connect density to floating and sinking. What additional information would you need to predict whether an object floats in a liquid?

E **Extend**
Design a two-test plan to distinguish two unlabeled solids using only physical properties.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Physical Properties of Matter

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. The amount of matter in a sample.

2. The amount of space a sample occupies.

3. Density = mass / volume.

4. Any three: solubility, conductivity, magnetism, plus mass, volume, density.

Sketch It - look for

Examples: balance for mass; graduated cylinder/ruler for volume; mass/volume calculation for density; dissolve test for solubility; circuit/tester for conductivity; magnet for magnetism.

Apply It

1. 2.7 g/mL.

2. Magnetism.

3. Electrical conductivity.

Reason from Evidence - look for

Measuring mass, volume, density, conductivity, magnetism, or solubility can identify a property without creating a new substance. Some tests may change form or location but not chemical identity.

ACE - Extend look-for

Accept reasonable plans using measurable properties such as density, magnetism, conductivity, or solubility, with clear expected evidence.

Suggested use	Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.